Quantitative Methods

Seminar: Measures and Charts

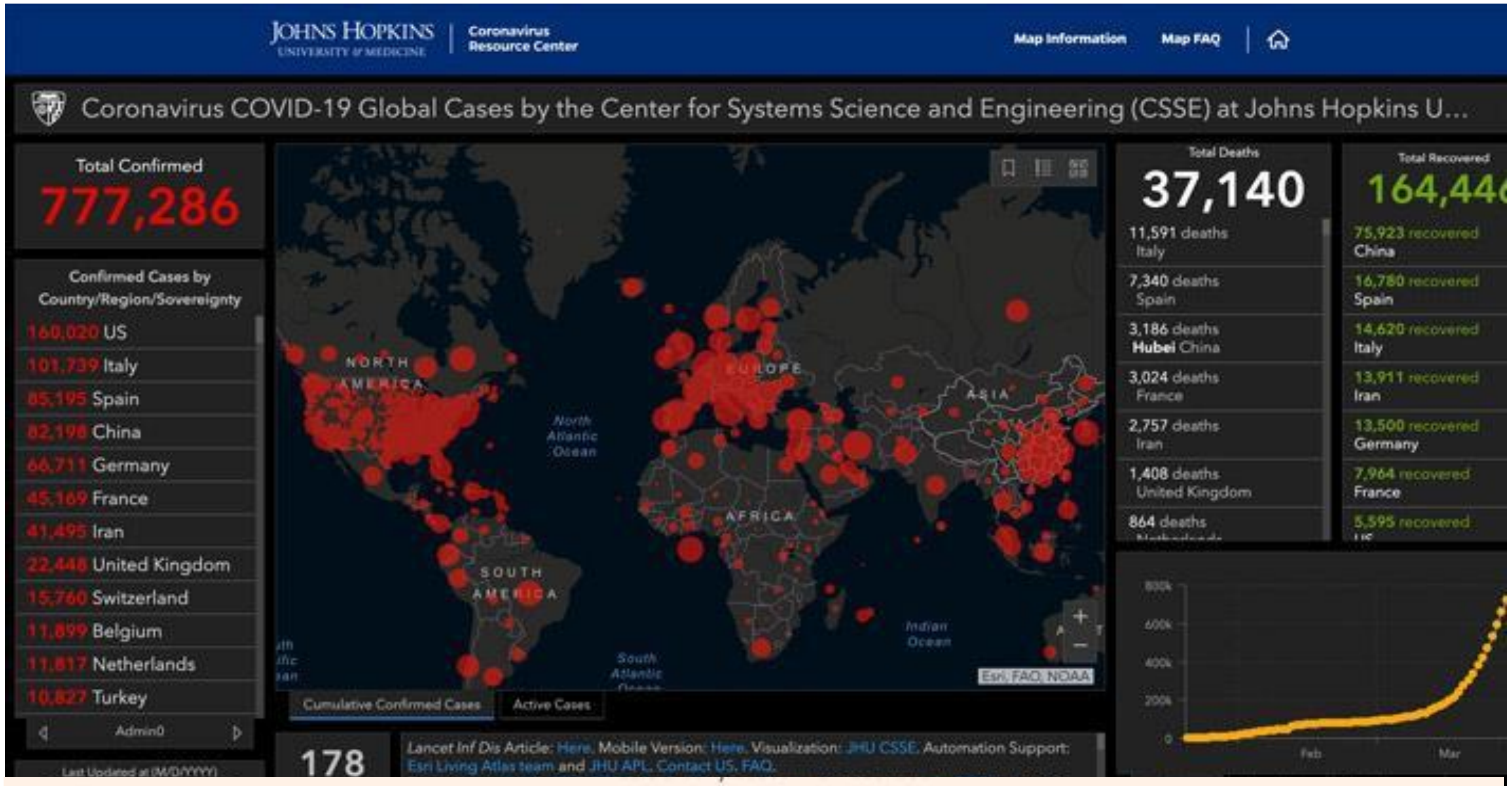
Recall from lectures…

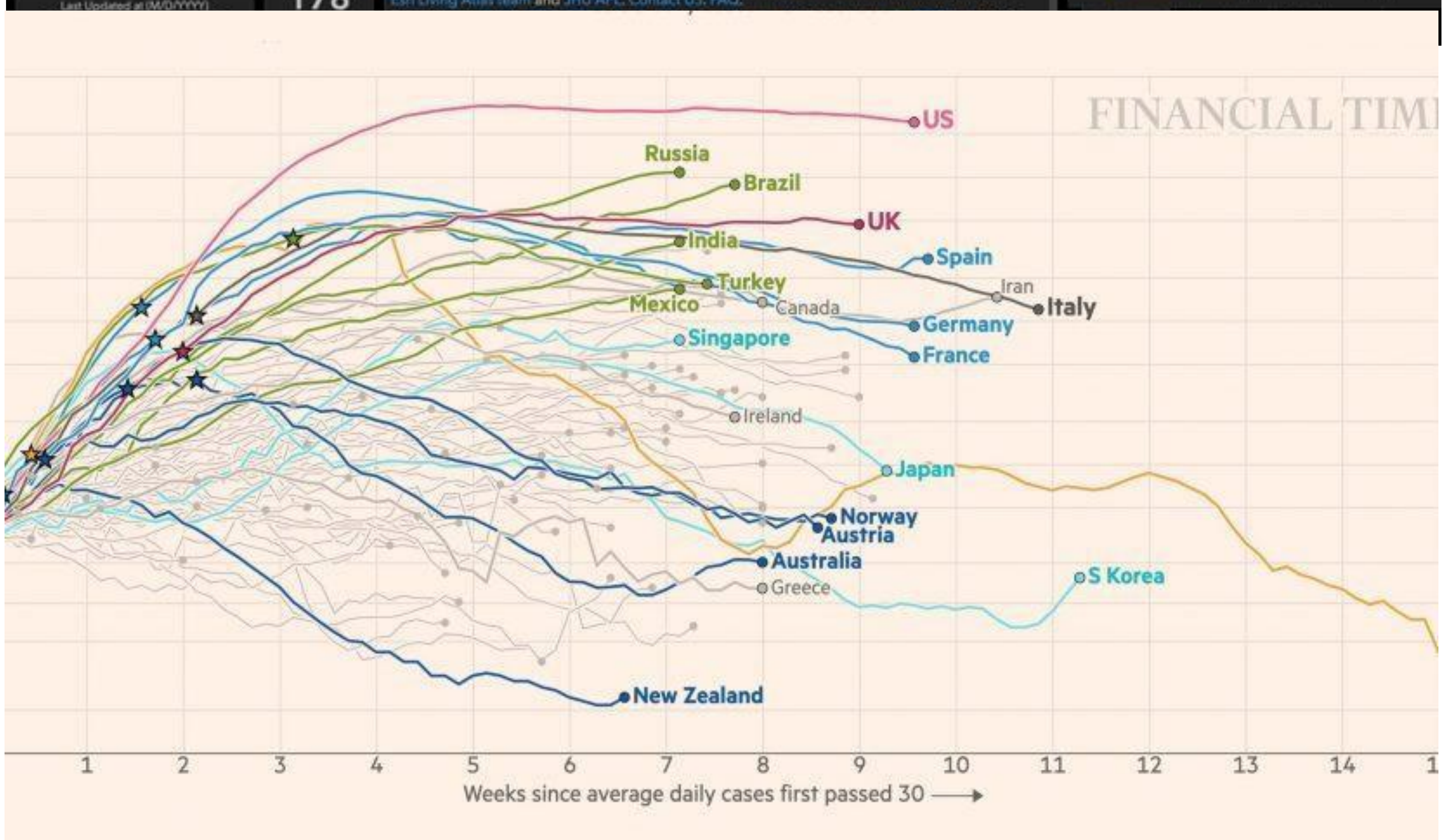
- We would like to extract information from datasets: describe the data
- Data description is extremely useful to understand the main trends
- Then we can continue with further analysis

This process is broadly used as baseline in academic research, policy making, consulting, businesses' strategic planning, and the press (i.e. Financial Times, the Economist etc.)

- ✓ Our aim is to understand how we can use some basic but extremely important concepts to analyse data

Data availability and basic analysis (such as mean-variance analysis) is easy but also crucial!





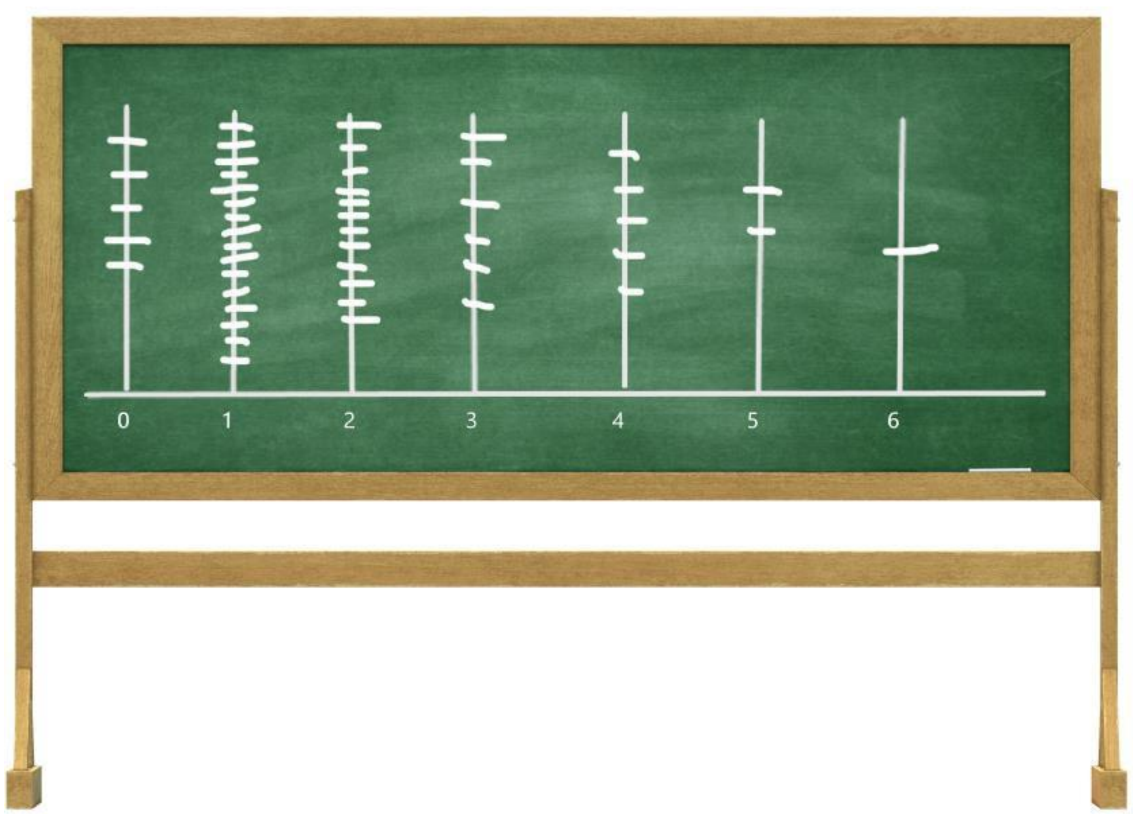
Especially in business management, data usually affect managers' decisions and strategy planning.

- **Question 1**

Calculate the median of:

Sales	Frequency
0	5
1	16
2	12
3	6
4	5
5	2
6	1

The median is the middle value, which is the value of the *(n+1)/2*th data point. Once we have frequencies, the number of datapoints is the sum of the frequencies.



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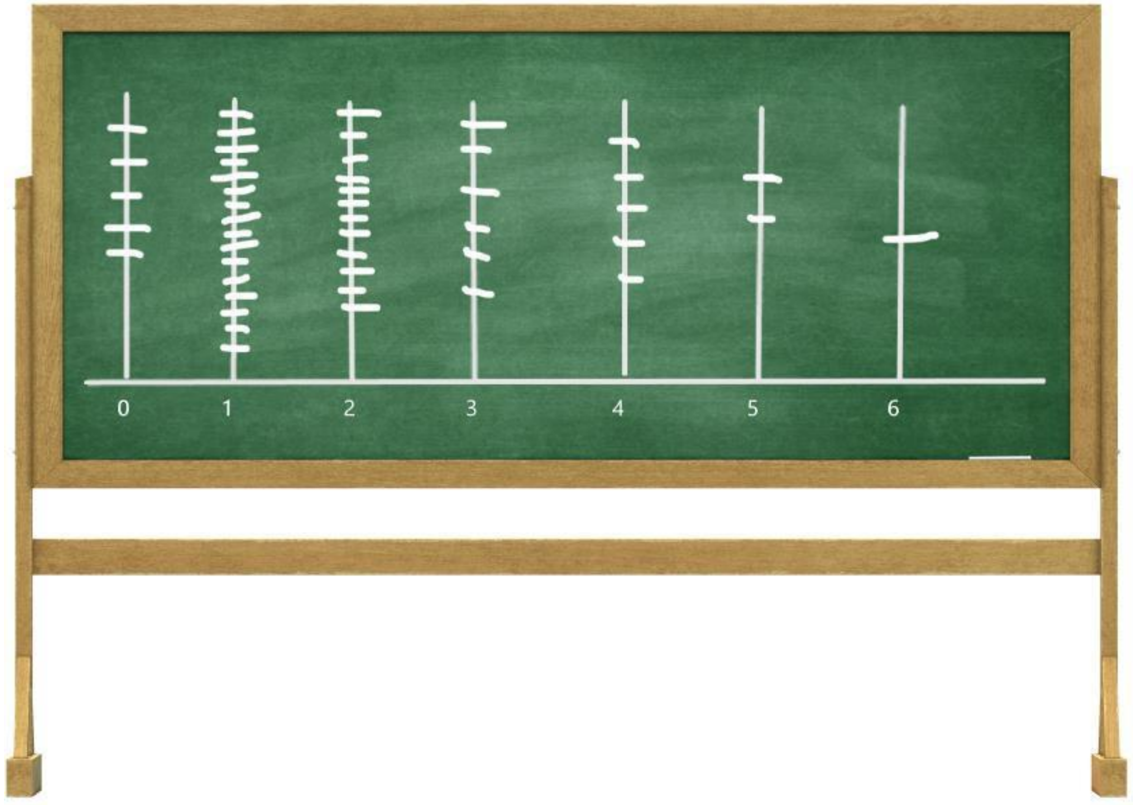
Let's count! How many datapoint do we have? 5+16+12+6+5+2+1 = 47.

Hence the median is the *(47+1)/2 = 24th* data point. What is the value of the 24th data point?



That is 2.

REMEMBER: the median and the mean are different concepts.

OTHER EXAMPLE: in a data set of {3, 13, 2, 34, 11, 26, 47}, the sorted order becomes {2, 3, 11, 13, 26, 34, 47}. The median is the number in the middle {2, 3, 11, **13**, 26, 34, 47}, that is 13.

Following the formula, we have 7 datapoints. Thus, the median is *(7+1)/2 = 4th point,* which is indeed 13.

- **Question 2**

The five-number summary of:

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Remember, the Five Number Summary is a set of descriptive statistics that provides information about a dataset. It consists of the five most important sample percentiles:

1. the minimum (smallest observation)
2. the lower quartile: $(n+1)/4^{th}$ datapoint
3. the median
4. the upper quartile: $3(n+1)/4^{th}$
5. the maximum (largest observation),

in that order.

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Hence,

1. the minimum: 0
2. the lower quartile: $(47+1)/4 = 12^{th}$, that is 1.
3. the median: 2
4. the upper quartile: $3(47+1)/4^{\text{th}} = 36^{\text{th}}$, that is 3
5. the maximum: 6

- **Question 3**

Calculate the Inter-Quartile Range of 1.79, 1.08, 1.51, 1.86, 1.28, 1.46, 1.38, 1.32, 1.09, 1.01 and 1.21.

The IQR is the upper quartile subtract the lower quartile. . The lower quartile (Q1) is the *(n+1)/4*th value, when the data points are in order, starting at the lowest. The upper quartile (Q3) is the *3(n+1)/4*th value.

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- Ordered data: 1.01, 1.08, 1.09, 1.21, 1.28, 1.32, 1.38, 1.46, 1.51, 1.79, 1.86. We have 11 datapoints.

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Q1 = *(11+1)/4 = 3*rd *point,* that is 1.09 Q3 = 3*(11+1)/4 = 9*th *point,* that is 1.51

IQR = Q3 – Q1 = 0.42

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Recall: The Five Number Summary is a set of the five most important sample percentiles:

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3. the median: $(n+1)/2^{\text{th}}$
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in that order.

Before we start, order the data!

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Before we start, order the data!

10, 12, 14, 14, 15, 16, 17, 18, 19, 19, 20. We have 11 datapoints.

1. the minimum: 10
2. the lower quartile: $(11+1)/4 = 3^{rd}$, that is 14.
3. the median: $(11+1)/2 = 6^{th}$, that is 16.
4. the upper quartile: $3(11+1)/4^{\text{th}} = 9^{\text{th}}$, that is 19.
5. the maximum: 20.

- **Question 5**

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To calculate them, we first order the data.

- **Question 5**

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Remember, the range is the difference between the biggest (maximum) and smallest (minimum) values.

To calculate them, we first order the data.

From Question 5 we have that the minimum is 10 and the maximum is 20.

Hence, range = 20 – 10 = 10

- **Question 6**

What is the standard deviation of the following data?

4, 93, 43, 93, 34.

$$\sigma = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(x_i - \mu)^2}.$$

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$$\sigma = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(x_i - \mu)^2}.$$

The mean is (4 + 93 + 43 + 93 + 34)/5 = 53.4

Thus,

$$\sigma^2 = \frac{\Sigma(x_i - \mu)^2}{N}$$
$$= \frac{(4 - 53.4)^2 + \ldots + (34 - 53.4)^2}{5}$$
$$= \frac{6061.2}{5}$$
$$= 1212.24$$
$$\sigma = \sqrt{1212.24}$$
$$= 34.82$$

Using the calculator: Remember, do this on the calculator. Mode-2 (STAT), put the data in, AC, shift-1, 4:VAR (or 5:VAR on the older model), 3:sigma x (or 3:x sigma n), =.

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Remember, to standardise, we take the value, subtract the mean and then divide by the standard deviation:

$Z = (X - \mu)/\sigma = (48 - 32)/ 9 = 1.78$

- **Question 8**

What's the mean of the following set of data?

x	frequency
0	9
8	10
12	25
18	12
21	10
27	6

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0	9
8	10
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21	10
27	6

$\mu = [(0*9) + (8*10) + (12*25) + (18*12) + (21*10) + (27*6)] / [9 + 10 + 25 + 12 + 10 + 6] = 13.44$

Using the Calculator: Remember you'll need Frequency turned on. (Shift-Mode (Setup), down, 3:STAT, 1:On.). For more guidance on using the calculator refer to the additional slides, i.e. Lecture 2 additional slides Measures and Charts.

- **Question 9**

Which of the following is true of a boxplot?

It demonstrates skew in the data

It allows the mode to be easily calculated

It shows the standard deviation of the data

It shows a trend over time

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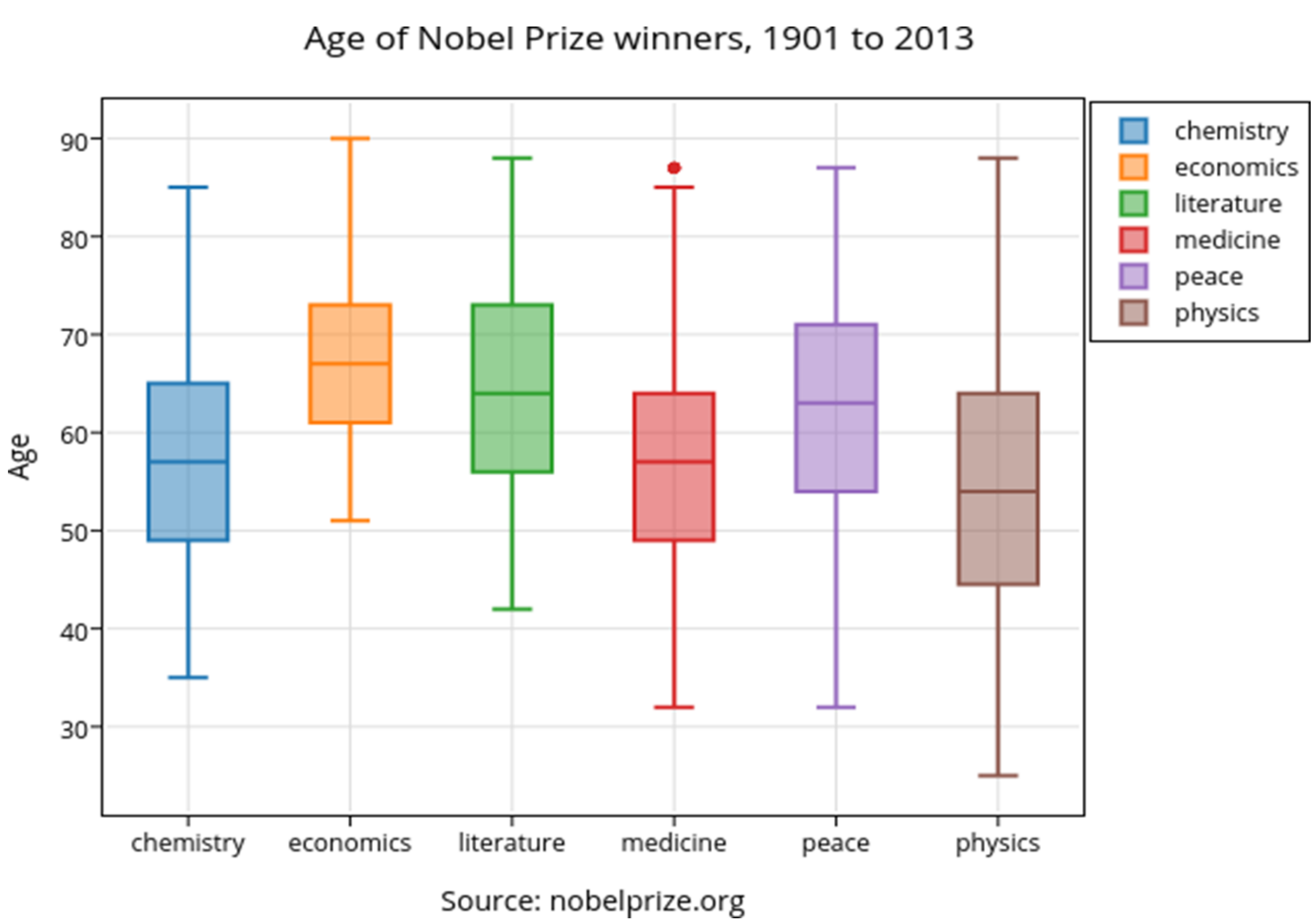

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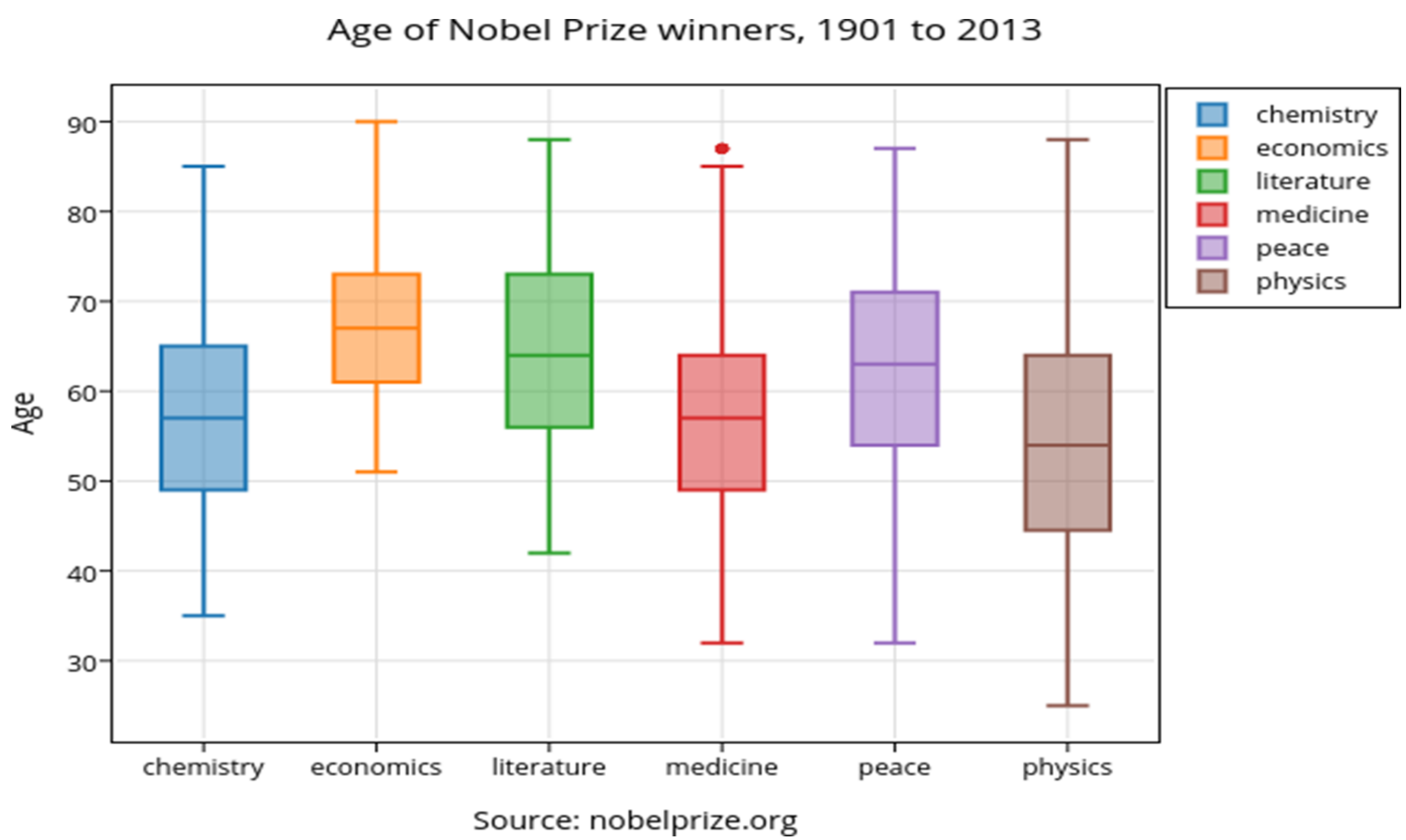

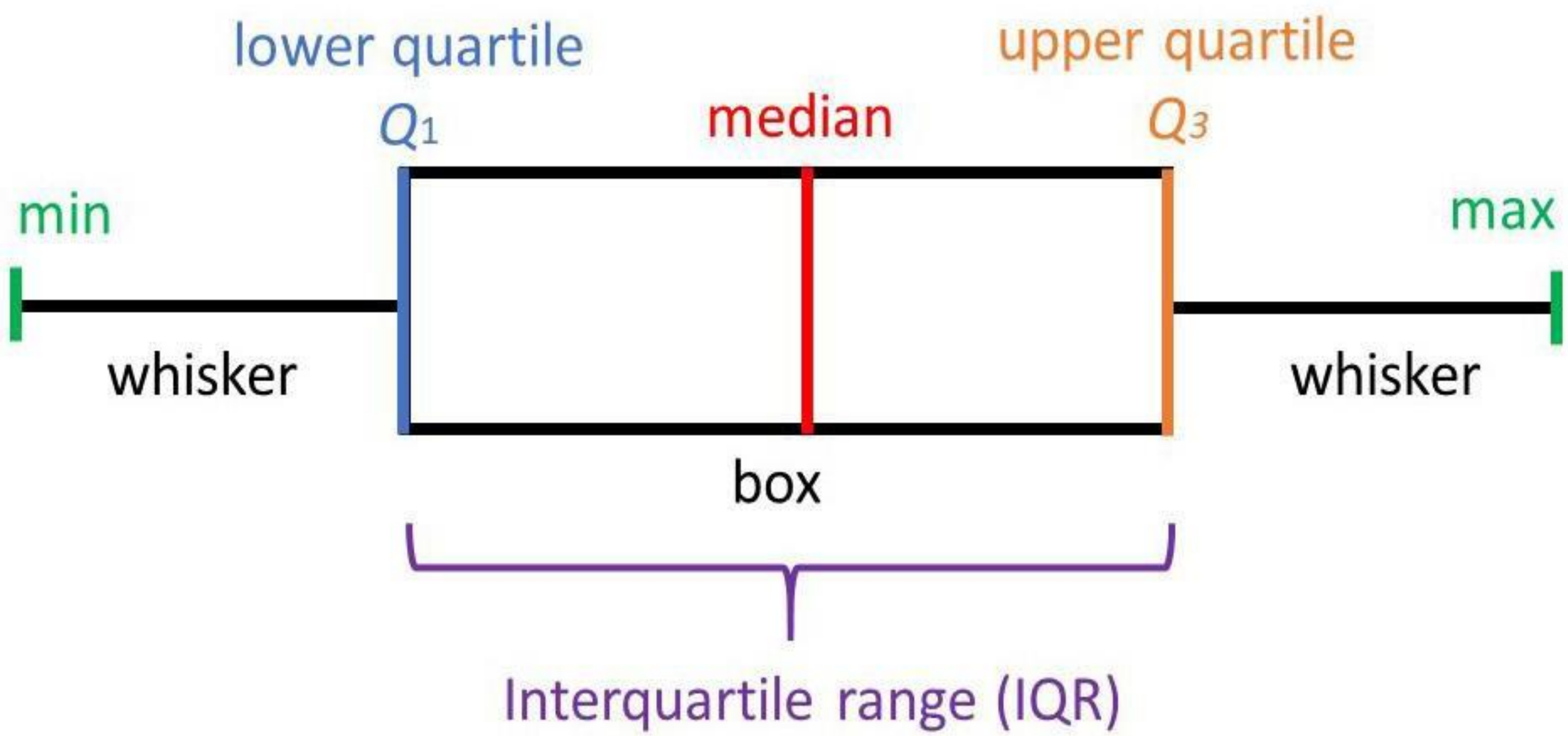

lower quartile
Q_1
median
upper quartile
Q_3
min
max
whisker
whisker
box
Interquartile range (IQR)

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What is the coefficient of variation of the following data:

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Coefficient of variation CV = Standard Deviation / Mean = σ / μ

The CV shows the rate of dispersion of data points in a data series around the mean.

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Coefficient of variation CV = Standard Deviation / Mean = σ / μ

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Calculate the standard deviation: $\sigma = 26.36$

Calculate the mean : $\mu = 48$

CV = 26.35 / 48 = 0.55

- **Question 11**

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Inter-Quartile Range.

Standard deviation.

Mode.

Range.

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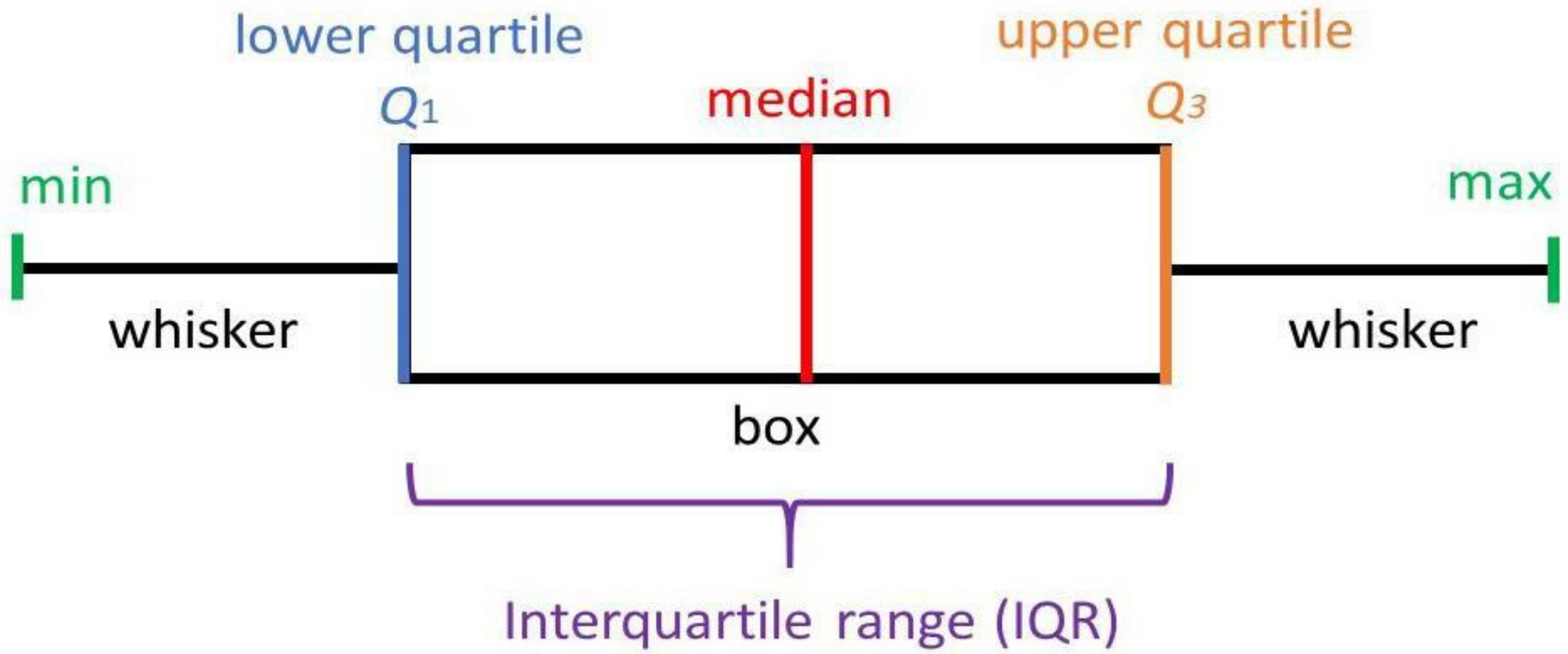

- **Question 12**

If the datapoints all decrease by 7, the inter-quartile range decreases by 7.

True or False ?

- **Question 12**

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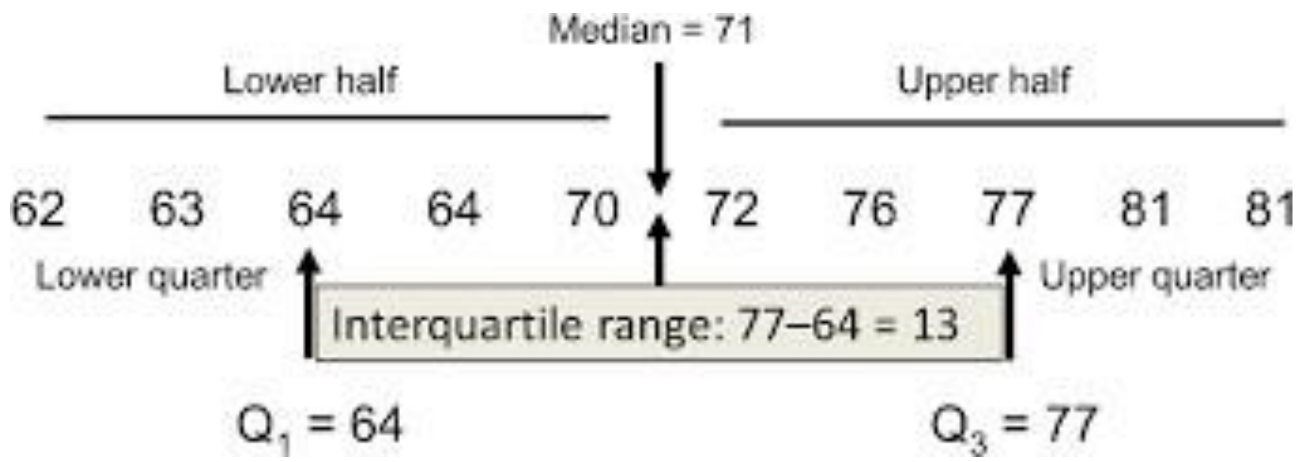
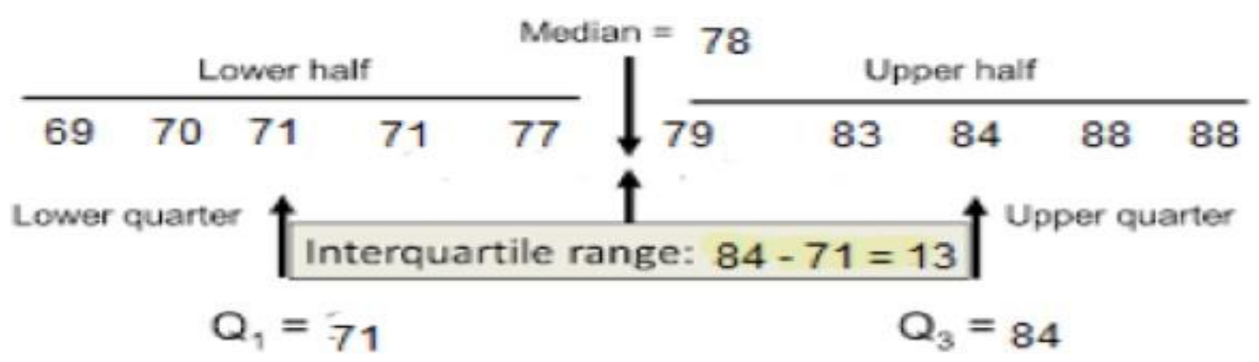
False!

the IQR is a measure of spread, not a measure of centrality. As such, it doesn't change as the dataset moves.





- Any questions?

Please try the **formative test of topic 1** – **Data, Descriptive Statistics and Charts - Formative Quiz** and let me know if something is not clear

- Thank you ☺